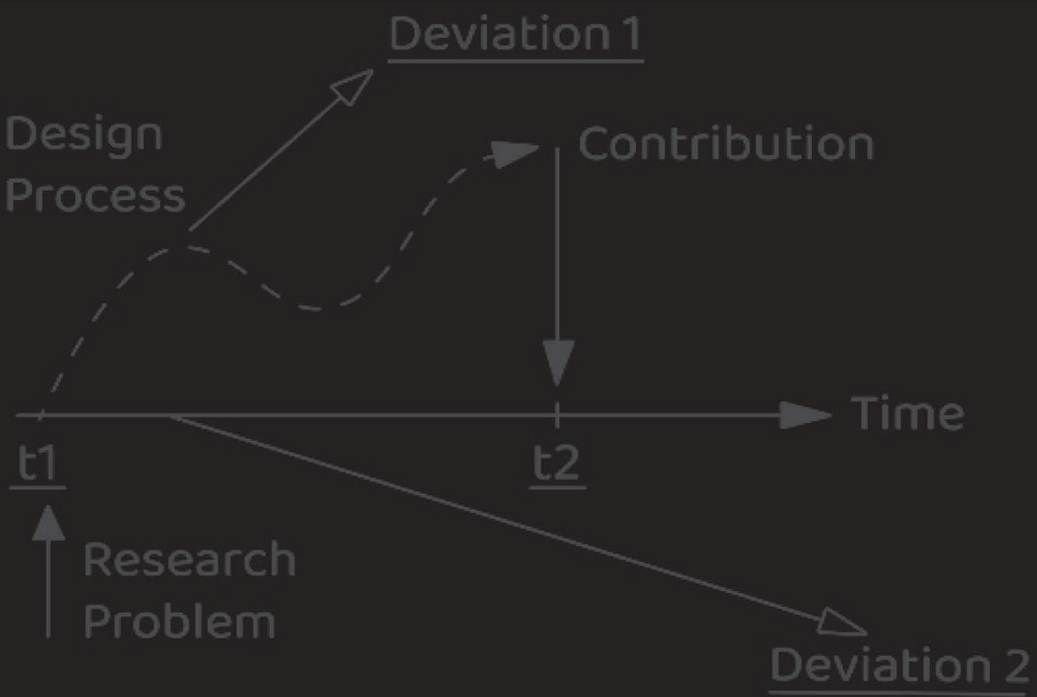
FAST AND SLOW

Methods for Timely HCI and Interaction Design Research



MIKAEL WIBERG

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Fast and Slow

Fast and Slow: Methods for Timely HCI and Interaction Design Research explores the dynamic interplay between rapid and reflective, or long-term vs. short-term research in human–computer interaction (HCI) and interaction design research. In an era where fast approaches to technical research are foregrounded, this book provides a critical examination of the temporalities at play in various research approaches – from long-term empirical studies to rapid prototyping. Ultimately, it asks the fundamental question of how to make timely research contributions and how to plan research projects to be timely in terms of impact.

Drawing inspiration from fast and slow thinking, design thinking, and our ever-changing world, this book contrasts "fast" approaches – such as "quick-and-dirty" ethnography, prototyping, and AI-driven automation – with "slow" methodologies that emphasize ethnographic studies, longitudinal research, and participatory design. By working across the fast and slow approaches to research, the book helps researchers and practitioners navigate the trade-offs between efficiency and depth, rapid results, and reflection. Ultimately, the focus of the book is on timely research contributions.

With a focus on research approaches, this book presents cases, methodological insights, and theoretical discussions that foreground the temporalities at play in HCI and interaction design research. It questions whether the rush to develop, iterate, and deploy can sometimes obscure critical insights about human behavior, emerging phenomena, ethical considerations, and long-term impact.

Whether you are an HCI researcher, UX practitioner, or technology strategist, *Fast and Slow: Methods for Timely HCI and Interaction Design Research* offers a fresh perspective on how to plan and carry out HCI and interaction design research – over time.



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Fast and Slow

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Mikael Wiberg



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Preface

Research takes time. And research contributions need to be timely – it is part of how a contribution is defined – it presents something new in relation to a research problem and in relation to an established body of related research. As such, research is relational and temporally situated. Research is also not just temporally situated, but also temporally motivated and restrained. Sometimes there is an urgent research problem that needs to be addressed, and sometimes there is just too little time when some longitudinal studies would be the preferred approach. Over time, research problems emerge, new methods are developed, and progress is made – through new studies, explorations, insights, discoveries, and new ways of seeing things. Through collective efforts – ranging from research collaborations, projects and research programs, to seminars, peer reviews, and presentations – research is always in motion and particular research projects are accordingly always situated and defined in relation to these collective movements – again, over time. And over time, new research problems are identified, old problems are revisited, and sometimes even old theories and conclusions are challenged. Sometimes there is a research breakthrough, where huge steps forward can be taken, and sometimes researchers struggle for decades with the same research problem. In short, this dynamics of how research is conducted is something that plays out over time. Given this, the pace of research is not constant, nor is it possible to talk about "one pace" or "one research process." Still, very little has been said about the different temporalities at play during the research process, the importance of being timely when selecting and addressing a research problem, and the importance of being timely to make a solid research contribution. This book is about these temporalities at play, about the "landscape of temporalities" that surround, influence, motivate, and restrain research – particularly in interaction design research.

If taking a step back, we can notice how the two concepts of "time" and "temporality" are often taken for granted. We use these words in our everyday language when we say "well, it takes time," "let's wait and see," or "let's wrap this up before it's too late," which speaks to how things evolve over time, or that it is about time to finish something. Somehow, there seems to be a window in time where a timely research contribution can be made. Still, although we use these terms in our everyday language, they are also quite complex concepts to explore. As I illustrate in this book, these notions of time, temporality, pace, rhythm, etc., can work as a vocabulary, perspective, and lens that allow us to see how things unfold in research processes – over time.

On the one hand, time can be thought of as a concept that we either experience or measure. For instance, how we might feel bored over longer periods with no activities, how we might feel when we are behind schedule, when we are about to miss a deadline, or when we lose track of time because we are immersed "in the moment." We can also use time as an indicator of how well we perform. We can formulate time plans, KPIs, GANTT charts, deadlines, etc., and through these time-based tools, we can estimate and then measure or evaluate how long some activities have taken, how far we are from a deadline, if we need to speed up, or how much time we have left. In short, time affects our behavior, activities, priorities, and how we make plans. But on a higher level of abstraction, we can also think about time and temporality as a fundamental construct that shapes our lives – from clocks, calendars, and to-do lists to deadlines, timetables, and ways of doing several things – one thing at a time, in a certain order over time, or at the same time, i.e., multitasking. Time, and how we relate to and organize ourselves around it, shapes our everyday lives, and as I illustrate in this book, it also shapes the research we do, and how we organize our research processes – explicitly or implicitly, no matter if we want it or not.

In the research areas of human–computer interaction (HCI) and interaction design (IxD), time and temporality play a crucial role in structuring research – over time. It influences the problems we address, the methods we choose, the technologies we focus on, the theories we pick, the timelines and deadlines we set, and how we adjust our research ambitions in relation to how much time we estimate that we have – explicitly in relation to project goals, work packages (WPs), and project duration, and more implicitly in relation to our understanding of how the research landscape is moving and changing, and if we will manage to make a timely

contribution to that movement. As I illustrate in this book, different paces are at play, with some processes moving quickly, while other activities and processes are moving more slowly – i.e., fast vs. slow research processes and approaches. As researchers, we are constantly making estimations of these paces, ranging from practical matters such as how urgent a research problem is, or how long a research project should be, to more overarching matters – such as if what we are studying is a new topic, a novel approach, a trend, or an escalating problem. Across all these cases, time is a central concern and a governing principle for any research project. No matter how you spin it, time will always be a factor.

In this book, I explore, contrast, and propose ways of unifying fast and slow approaches to interaction design research. Related to these perspectives on "fast" vs. "slow" approaches to design research, I have of course been inspired and influenced by the pioneering book *Thinking, Fast and Slow* by the Nobel Prize winner, Professor Daniel Kahneman (Kahneman, 2011) – a book that has already been cited more than 53,000 times. I have also been inspired by the growing body of research on what has been labeled "the slow science movement": see for instance the book *The Slow Professor* by Berg and Seeber (2016), and the book *Another Science Is Possible: A Manifesto for Slow Science* by Isabelle Stengers (2016). These books examine an alternative, slower temporality for conducting research, and it is interesting to relate these alternative approaches to the temporalities at play in interaction design research. I have also been inspired by the growing body of related research that looks into fast or slow approaches to research or work that considers the temporalities at play when planning or conducting short or longitudinal research, for instance, the work by Sauer (2018) on fast and slow moral thinking, and the work by Kannengiesser and Gero (2019). On fast and slow design thinking see Earley (2017) and Keulemans (2021) and on fast and slow design see Frohlich (2015). I have also been inspired by work on fast and slow innovation: Gasparin et al. (2020) on slow design-driven innovation and Goldsworthy et al. (2018) on "circular speeds" and fast and slow design approaches. I have also been inspired by McCracken's (1988) book *The Long Interview*, Croskerry et al. (2014) on fast and slow decisions, Cal Newport's book *Deep Work* (Newport, 2019), and Dorie Clark's book *The Long Game: How to Be a Long-Term Thinker in a Short-Term World* (Clark, 2021). Finally, I have also been inspired by more general literature on time, temporality, and the experience of time, for instance, the book *Fast and Slow: Design and the Experience of Time* by Hayward (2016).

Understanding and foregrounding the different temporal paces at play in HCI and interaction design research is important because it allows us to navigate and make plans in a rapidly changing research landscape. At the current moment, there are several drivers that push for fast approaches in our field, including new and emerging technologies (at the current moment in time, this includes, for instance AI – artificial intelligence, and IoT – the Internet of Things), research funding, call for papers, and bibliometrics. On the other hand, there are calls for slow approaches to address complex and hard-to-solve research problems. In this context, fast approaches have often been seen as "relevant" and well-aligned with tech paradigm moves and the buzzword-driven landscape surrounding digital technology research and development. On the other hand, slow approaches have typically been associated with "rigor" and the formulation of new paths, paradigms, theories, research directions, and new orientations and movements in a field of research.

Traditionally, HCI and IxD research have focused on fast solutions (most recently discussed as the "solutionism" movement in HCI. For a discussion on this, see for instance: Cunningham et al. (2023), Adamu (2023), Aanestad (2023), Blythe et al. (2016), and Oulasvirta and Hornbæk (2016). Along this line, we find a set of approaches for going "fast" in the research process, including methods like brainstorming (see e.g., Wilson, 2013; Sutton & Hargadon, 1996); Besant, 2016; and Putman & Paulus, 2009), design briefs (Ryd, 2004), quick and dirty ethnography (Twidale et al., 2014), and rapid prototyping. However, there has been a recent turn in the field of HCI toward "big debates" including topics such as race, gender, climate, sustainability, and social justice, to just mention a few of these more complex topics. In short, a set of issues and problems that cannot really be quickly solved, and things that cannot just be addressed once and then call it solved. On the contrary, these issues call for long-term and slow changes, and these topics even call for fundamental and systemic shifts. Such shifts take time.

Ultimately, as I illustrate in this book, the choice between, or the combination of, fast and slow methodological approaches in HCI and interaction design research is a matter of paying attention to the urgency, complexity, and scale of the research problem – in relation to how a particular field of research develops – over time. Large-scale research problems require collective efforts and coordination, often involving large interdisciplinary research teams and long-term funding. This often necessitates slower, more rigorous approaches to understanding and addressing large-scale

challenges. On the other hand, research contributions are typically made through the documentation and publication of timely studies in relation to how a research field develops over time. Accordingly, it becomes a practical and everyday challenge for any researcher who wants to make a difference – to simultaneously address an urgent and large enough research problem, while managing to break it down into smaller (and shorter) studies and aim for timely contributions.

So, how do you as an individual researcher navigate this ever-changing temporal landscape? How do you plan your research projects so that you can make timely contributions? And ultimately, how do you contribute to where the field of research should go next, or where it needs to be, in say 10 years from now?

This is what this book is about.

–"Go fast, Go slow" – Enjoy!

About the Author

Mikael Wiberg is Full Professor in Informatics at Umeå University, Sweden. He has held positions as Full Professor in Interaction Design at Chalmers University of Technology and as Chaired Professor in HCI at Uppsala University. Wiberg's research is within the area of HCI/interaction design research, and he has published his research in top journals including *Design Issues*, *Design Studies*, *International Journal of Design*, *ToCHI – ACM Transactions on Computer–Human Interaction*, *the Human–Computer Interaction* journal, *Interacting with Computers*. He is also co-EIC (Editor-in-Chief) for *ACM Interactions*.



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CHAPTER 1

Fast and Slow Design Research

INTRODUCTION

How we step-by-step move forward in interaction design research, and how we motivate and describe these steps, is a matter of method. And, at what pace we move forward through these steps is a matter of the temporalities at play – no matter if we move rapidly through the research process, or if we are slowly and steadily moving forward.

In this book, we examine the temporalities at play in interaction design research. We do this through an examination of how design ambitions and research efforts come together in design research. This examination is grounded in an analytical approach to how design research has developed over the last decade. The book is also rooted in my own experiences from doing design-oriented research over the past 25 years – in particular in the research areas of HCI (human–computer interaction) and interaction design research. While there is a growing body of research on design, design methods, and research approaches to design [for instance, RtD – Research through Design (see e.g., Zimmerman & Forlizzi, 2014; Bardzell et al., 2016; Godin & Zahedi 2014; and Giaccardi, 2019)] less is said about *the temporalities underpinning and influencing interaction design research processes*. In this book, I suggest that an understanding of these temporalities at play is of essential value and importance for a number of reasons: 1) for the strategic planning of interaction design research projects, 2) to

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understand the relation between design and research processes, and how these come together in design research, and 3) to understand how to align basic and applied design research agendas in practice. Taken together, these three aspects are crucial for doing timely research, and accordingly, it is essential for making timely research contributions in the area of HCI/interaction design research.

On an overarching level, this book is about the processes, rhythms, and paces of interaction design research. It is about the *practice of doing* timely research, and it is about *research strategies* for how to aim for long-lasting impact – through interaction design research. It is about how to combine a focus on the most pressing issues and challenges, while aiming for long-term progress and long-lasting impact. Historically, HCI and interaction design research have been very future-oriented, design-oriented (Wiberg, 2014; Wiberg & Stolterman, 2014), and problem-solution-oriented (Oulasvirta & Hornbæk, 2016). Solutions and novelty have been criteria to measure if a research project is producing valuable results – or not. However, this situation is now changing, and there are even papers that criticize HCI for pushing a "solutionism" agenda in favor of more long-lasting, or more long-term impact. In short, it is about going fast – and finding solutions to problems, or going slow – and working on more complex (and even wicked) research problems.

There are of course several books already on design methods and design thinking, and there are books available on particular areas of design practice, for instance, books on interaction design (e.g., Lowgren & Stolterman, 2007). Still, there is a lack of literature that specifically addresses the temporalities at play in interaction design research. This book serves as an important contribution to this need. If we look at the existing strand of literature there are a set of books that are of particular relevance here, although none of these books focuses on the temporalities of design research. For instance, the book *The Design Way* by Nelson and Stolterman (2014) provides an in-depth perspective on design, design theory, and design as an approach. This is an important book as it provides a foundation for understanding the nature of design and design processes. In close relation to this book, I would also like to acknowledge the book *Making Design Theory* (MIT Press) by Redstrom (2017). This book is important as it describes how theorizing can be made in design research, how applied and basic research is interrelated in design research, and how design researchers can work along "programs" in their research efforts.

In addition, we have the book *Transmissions* (MIT Press) by Jungnickel (2020). This book looks at the outcomes, presentations, and dissemination of design research. According to the author of this book, "Transmission is the research moment when invention meets dissemination—the tactical combination of making (how theory, methods, and data shape research) and communicating (how research is shown and shared)." This is a book of great importance and value for understanding the processes surrounding design research, in particular toward the end of a research process where this transmission is not just about showing and sharing research results, but about presenting these results in a timely manner. In short, I think this book adds to these other important books by providing a particular focus on the temporalities surrounding and underpinning interaction design research processes, and how to align those internal research processes with what is happening in practice and what is happening in our surrounding society – at a given moment in time.

We can also notice here how this book on *timely interaction design research* is well-aligned with a number of books that point to the dynamics of our everyday, and accordingly how we need very dynamic design research processes that are well-suited for and capable of following and contributing to this ever-changing landscape of our everyday. Here, I would like to mention the classic book *Beyond the Stable State* by Donald Schön (1971), the book *Liquid Modernity* by Zygmunt Bauman (2013), and the book *The Necessity of Friction* by Nordal Åkerman (2018). These three books are all about this ever-changing dynamics in our everyday life. On the level of the individual, and how we need to cope with an ever-changing everyday full of challenges, changes, and interruptions, I am also thinking about the book *The Distracted Mind* by Adam Gazzaley and Larry D. Rosen (2016). Still, and while all of these related books examine and foreground the dynamics of our ever-changing everyday, none of these books address this from the perspective of how to plan and carry out interaction design research that is timely in relation to such unstable, uncertain, and dynamic conditions, circumstances, and surroundings. This is where this book on timely approaches to design research contributes with a perspective on how to plan and work with fast and slow approaches to interaction design research in relation to this dynamic object of study. In addition to this, if we look at how this book might contribute more broadly to the field of interaction design research, I would say that this book brings together these perspectives offered by the existing literature (i.e., the books on

design/design theory, the books on the dynamics and ever-changing nature of our contemporary society, and books that have highlighted the importance of alternative temporalities for doing research).

By bringing these perspectives together, I see this book as something that hopefully offers a set of important contributions in relation to the growing body of interaction design research that specifically addresses ways of approaching design research (e.g., design research methods) and ways of theorizing design research. In addition to this, I think that this book might serve as a thought piece for design researchers interested in reflecting upon the temporalities, rhythms, and paces that form, influence, push, or even restrict research projects. It might also serve as a guiding framework that can make design researchers better – both in terms of planning new projects (a process perspective) and in terms of making timely contributions (an outcome perspective). Finally, I hope that this book can provide some guidance on thinking about interaction design research as a matter of work that is oriented toward the formulation, articulation, and realization of *prospects for alternative futures*.

While my own research over the past 25 years has mostly been in the area of interaction design/HCI (human–computer interaction), I think that this book might be relevant for almost anyone interested in design research approaches and ways of doing timely research through design. As such, I see the audience of this book being designers, design researchers, HCI and design students, PhD students in HCI/interaction design, interaction designers, researchers in HCI – or in any other area where design, proposals, recommendations, or implications are part of the method, approach, or foundation for doing research and for moving forward through timely contributions that make our world a better place – for all of us!

Some Initial and Fundamental Questions – on Fast and Slow Interaction Design Research

When do we design? *What* does it mean to do timely design research? And accordingly, *how* should we plan our interaction design research projects? Indeed, these are concerns for the temporalities of interaction design research. When we plan and carry out interaction design research projects, we need to account for and deal with the temporalities of research processes. We need to do our projects on time, in time, and within deadlines, and we need to be timely with our results and contributions. Still, when reviewing the existing literature on interaction design research,

very little is said about these temporalities. So, what are *the rhythms of design* and the *paces of research*? And how can these two come together in interaction design research projects? On an overarching level, this book is about the processes and paces of interaction design research. It is about doing timely research, and how to aim for long-lasting impact. It is about how to combine a focus on the most pressing issues and challenges while still aiming for long-term progress and impact.

In this book, interaction design research is examined from a temporal perspective. It includes both the planning of HCI/interaction design research projects, as well as with internal and external factors influencing the temporality of these processes. For instance, we might need to *move fast* to find solutions to pressing problems, or we might need to be first to contribute to a fast-growing strand of research. In other processes, we might need to dig deep and *move slowly* to fully understand and analyze a particular setting. We might also need to be future-oriented in the planning of our research projects, ensuring that our contributions toward the end of the project will be relevant at a future moment in time. Certainly, interaction design research moves forward at multiple paces, making it is accordingly a relational concern to ensure that the things we do now will be relevant not just in the present, but also at later stages and for longer periods of time.

In addressing *the rhythms and paces of interaction design research*, I will, in this book, propose and use the notions of "fast" and "slow" as a vocabulary that allows for an examination of these rhythms and paces of design research. Here, I define fast and slow as relational notions. A research process is *fast* if it is more rapid than how the research problem changes or develops. If you are fast, you can address it while there still is time, or while the research problem still exists or is relevant. Likewise, being *slow* is also relational. It can be that you are too slow, so you arrive at a solution but it is too late. It can also mean that you decide not to address the urgent research problems, but instead try to tackle or at least understand the more fundamental or underlying problem. This will take more time and accordingly lead to a longer and as such slower research process. Accordingly, I will rely on an understanding of "fast" as being about things that move forward rapidly, in quick succession, or ahead of an estimated time, deadline, plan, or schedule. Further, I will view "slow" as a different pace, although not necessarily a negative one (e.g., "being delayed"). Instead, I approach "slow" as a process that has qualities that might be necessary for moving forward at a rapid pace, through rapid progress and

actions. Here, slow processes that have involved reflections and activities that have laid a solid foundation might provide the grounds necessary for dealing with urgent research problems – at a later stage in time.

In our language, "fast" is typically associated with "action," "achievements," and "progress," whereas "slow" is typically associated with "reflection," "analytical," and "systematic" approaches. In interaction design research processes, we typically combine these two. We seek, we do, we iterate, we evaluate – we *act*, and we *reflect*. This is one way through which the fast and slow approaches come together in interaction design research practice. But we also have similar paces in research. We might be stuck with a research problem for a very long time, and suddenly we have a breakthrough. It is hard to predict *when* research demands further analysis, and *when* there will be a sudden shift that enables a rapid movement forward. Again, this is an example of the temporalities at play. So what do I then mean when we talk about paces and rhythms in this context? Well, *pace* is the overarching speed of the research project. Again, it is relational in many ways. The pace might be high in comparison with the research plan, or the pace might be slow because of challenges with getting access to data, etc. The notion of rhythm is related to this pace. On an overarching level, the pace might be slow, but there might be recurring moments with a higher pace. For instance, if the plan is to run a set of workshops as part of a project, then these workshops might run smoothly and according to the project plan, although the project as a whole is moving forward at a quite slow pace. As a researcher, it is accordingly important to not only understand the overarching pace of a project, but also to have a good understanding of fast vs slow activities, temporal bottlenecks, and time-consuming moments. With such knowledge, it is easier to plan research projects over time and be in a good position to make timely research contributions.

On a related note, it should be acknowledged here that I do not see "fast" as being sloppy or not being rigorous enough, and I do not see "slow" as a matter of being lazy, delayed, or too late. Instead, the two temporalities of *fast* and *slow have* their unique qualities, and in order to do timely design research, it is essential to understand how these different paces – these different temporalities of design research – are interrelated and interdependent.

On Timely, Recurring, and Eternal Research Questions

Beyond these two notions of moving fast and moving slow, there are also different types of research questions. Some questions are formulated

to address urgent problems, while other questions are formulated to address slower or less urgent problems. But there are also questions that will probably stay with us for a very long time, for instance, questions concerning sustainability. As you will see in this book, I will move from thinking about fast vs. slow research to discussing this in relation to basic and applied research. Here, the idea is that applied research is probably more suitable for a fast approach, and vice versa with basic research and a slower approach to research. But beyond these two alternatives, there are probably research questions that will stay with us – some kind of eternal research questions. Here, I will introduce this idea and then return to this topic toward the end of this book.

For sure, every field of research has its own set of "eternal," or at least recurring research questions. In our field, the field of human–computer interaction/interaction design research, we seem to revisit the question of how to keep up with the pace of technological developments, and sometimes we even seek to add to the forefront of technological advancements, while paying attention to the societal adoption of new digital technologies. At the same time, we try to make sure that we contribute to something more fundamental or long-lasting, i.e., to generate new knowledge and new theories about these processes.

As a field of research, we seem to struggle with how to conduct research that is *fast enough* to keep up with the current pace, and at the same time, engage in interaction design *thoughtfully* (Lowgren & Stolterman, 2007) *and thoroughly enough* to meet the scientific criteria of rigor. As a field of research, we need to contribute with concrete results and solutions (new gadgets, tools, and digital services), something that has sometimes been referred to in the literature as "ultimate particulars" (Nelson & Stolterman, 2014), while at the same time making sure that we generate abstractions, generalizations, and that we theorize (Stolterman & Wiberg 2010**;** Wiberg & Stolterman, 2019; Wiberg, 2018). It is as if we both want fast and concrete results that we can show, demo, and exemplify with, while at the same time coming back to the fact that we also want to make conceptual advancements in our field of research.

In adjusting our research agenda in relation to the pace of technological developments and changes in social practice, our field has been occupied with the latest technologies and with what's "new and novel" (Wiberg & Stolterman, 2014). We have also developed a keen eye toward innovation, novelty, and the next big thing. When we report our research findings, we nowadays typically strive to include "implications for design" (Dourish,

(2006), and we are excited and explicit about our design-oriented and future-oriented research agenda. We want to make an impact, and we want our research to matter – right now, and right away! To be relevant, timely, theoretically grounded, and future-oriented seem to be four central cornerstones for our field.

From a temporal perspective, we do our research "here and now," but at the same time we want it to be about the coming technological shift, the next big thing, e.g., next wave of computing, or future use scenarios. We want to understand and describe the present, but we also want to be normative and suggest where we should go next. In a sense, we are a forward-looking field rather than a field that only seeks to understand the current moment, and we seem to be occupied with new technologies and what futures these new technologies can enable for us (Bell & Dourish 2007). We are not only future-oriented, but in this strive, we are interested in the new and novel (Wiberg & Stolterman, 2014). For instance, HCI's most established conference, the ACM CHI conference, is explicit on how researchers should present their work in terms of its novelty. There is an embedded assumption here that we should not just focus on the current situation or the most recent technologies, but we should also draw conclusions about the next steps forward. We should be future-oriented, and we should seek to target, or even establish, the next phase or the next "wave" of computing. Indeed, our field has, over the last 20 years, been expanding across such waves, and in multiple new directions (for further discussion on these waves see e.g., Bødker, 2006, 2015; Kaptelinin et al., 2003, April; Goh et al., 2017; Frauenberger, 2019; Ashby et al., 2019, November; Homewood et al., 2020, July; Lopes, 2021; and Wiberg & Stolterman 2021). In addition, the *ACM Interactions* magazine has just recently launched a new publication format called "waves" devoted to articles that report on new and emerging topics and strands of research in our field.

It needs to be stated here that we are not only future-oriented, but we are also a design-oriented field of research. We are in that sense breakthrough-oriented, and we are expected to deliver timely research contribution – the CHI conference even has a submission category for "late-breaking results" as to ensure that the most recent results will reach out as soon as possible. At the same time, we are design-oriented, both in our approach and in our strive, as well as in terms of theoretical advancements. We want to expand on design research as a body of research, and we want to make theoretical contributions to the field of HCI. With this as part of our research agenda, we have pressure on us to systematically develop our theories, be critical,

contribute to the existing body of research, and generate insights and new knowledge with long-lasting implications. In short, we are expected to be both *fast* and *slow* in how we conduct our research. This is important, as these expectations shape how we conduct research in our field.

In this book, I focus on these different goals and temporalities of HCI and interaction design research. I discuss how to navigate the different paces surrounding our research projects, and how to do interaction design research that is both timely and long-lasting. For sure, our field has managed to do this and we actually have experience from doing it with every interaction design research project we conduct. Still, very little has been written about it from a research process perspective. As such, this book contributes to the existing literature on methods in HCI (human–computer interaction) and interaction design research, and it provides a reflective account that will hopefully help new scholars entering our field, particularly in understanding how to combine both fast and slow activities in research plans to be both timely and relevant, as well as critical, thoughtful, and grounded. This is at least the intention of this book.

TAKING A STEP BACK TO REFLECT – INTERACTION DESIGN RESEARCH, AND ITS FAST AND SLOW METHODS

On an overarching level, one can say that interaction design research is a field concerned with people, activities, context, and technologies (sometimes referred to as the "PACT" approach Benyon, (2014)). One might also say that it is about the design and evaluation of interactive products and systems, such as software applications, websites, and smart devices (see e.g., Wania et al., 2006; and MacDonald & Atwood 2013). For sure, usability testing has played a central role in the field of human–computer interaction, where usability scales such as the one proposed by Brooke (1996) has been frequently applied and referred to. At the current moment, this proposed usability scale has more than 20,000 citations. As technology continues to permeate our everyday lives, the importance of good interaction design has not only grown, as it plays a critical role in determining the usability, effectiveness, and satisfaction of these products and systems, but it has become fundamental since these technologies are now so entangled with our everyday lives that any attempt to separate the two is meaningless, if not impossible. Some researchers, e.g., Frauenberger (2019) has referred to this as the fourth wave of human–computer interaction – or entanglements – a stage where we are not only using and surrounding ourselves with these digital systems, but where these systems also to a large extent

define who we are, what we can (and cannot) do, how we understand and relate to the world around us, and how others perceive and interact with us. In short, we are existentially entangled with these digital systems in the fourth wave of HCI.

If now taking a step back, we should acknowledge how design research, as the broader research domain surrounding interaction design, is a field devoted to a systematic and rigorous approach to understanding the needs, motivations, and behaviors of people in order to inform the design of products, services, and experiences (Hartson, 1998). It is typically described as an interdisciplinary field (Blackwell (2015); Mackay & Fayard (1997); and Rozanski & Haake (2003) that combines elements of human-centered design, social sciences, and engineering to create a holistic understanding of the design context. The goal of design research has been described as generating insights that inform and inspire the design process, leading to innovative and effective solutions (Packer and Keates, 2024). Further on, it has typically been described as being done through an iterative design process that involves gathering and analyzing data from a variety of sources, including interviews, observations, surveys, and experiments (Dow et al., 2005; Myers, 1994). There are also many different methods of design research, including qualitative and quantitative approaches, that can be used to gather data from various stakeholders. These include user interviews, focus groups, usability testing, and field studies, among others. One key aspect of design research is the ability to identify and prioritize the needs of the target audience. This requires an empathic understanding of the users, their values, their needs and activities, and their motivations. It also requires an understanding of the broader context in which the design will be used, including cultural, social, and environmental factors. From the early ideation stage through the final evaluation of a product or service, design research can be used at any point in the design process. It is a crucial tool for designers who want to build solutions with effect and meaning that cater to the demands of their users.

Design research can also be part of a traditional design process (Jones, 1992), where it assists professional designers in the early stages of the process by defining the issue they are seeking to address and locating the important parties involved (Svetel et al., 2018; Collins et al., 2016; Zimmerman et al., 2007). Insights about the users and their needs can also be uncovered, which can guide the creation of personas (Chang et al., 2008) and user journeys (Benford et al., 2009; Bascur et al., 2019; Tan, 2023). Design research can also be used to evaluate and validate concepts

and prototypes as the design process develops. This can be accomplished using techniques like focus groups (Adams & Cox, 2008; Rosenbaum et al., 2002), user interviews (Blandford et al., 2016; Raita, 2012; Kuniavsky, 2009), and usability testing (see for instance Nielsen, 1992, 1994; Lewis, 2012; Shneiderman, 2000, and Issa & Isaias, 2022). Designers can also evaluate and enhance their designs with the help of the comments and information acquired through these methods and techniques.

In the final stages of the design process, design research can be used to evaluate the effectiveness and impact of the final product or service. This can involve gathering data on usage patterns, user satisfaction, and the overall user experience. Overall, the role of design research in the design process is to provide a solid foundation of knowledge about the users and their needs, which can inform and inspire the design process, leading to innovative and effective solutions.

Related to these contextual factors, the existing strand of research methods, and this recurring focus on users and users' needs, we should also acknowledge a recent movement in the field of design research that seeks to decentralize the human in these processes. Over the last few years, this has been labeled more-than-human centered design (Coulton & Lindley, 2019; Wakkary, 2021; Camocini & Vergani, 2021; Poikolainen et al., 2022; Giaccardi et al., 2024), and by de-centralizing the human this approach suggests that we need to focus on larger (eco-)systems to tackle global sustainability challenges. In a sense, this recent shift from *human-centered* to *more-than-human centered design* research serves as a good example of this combination of fast and slow movements in our field of research. On the one hand, there have been lots of rapid research projects conducted where people have looked for human-centered solutions to various problems. At the same time, there has been a slower movement underneath the fast-paced research, a movement where researchers have started to question if these solutions should only serve human needs, and the environmental footprint associated with pushing a human-centered research agenda. With the turn to more-than-human approaches, it is on one level a slow turn in the field, where researchers are seeking an alternative foundation for doing and evaluating research outcomes, but it also comes with an additional set of more rapid research where researchers are then conducting more-than-human centered interaction design research projects in attempts to explore what this turn could mean in practice.

Design research can also be used to assess the success and impact of the finished product or service in the last stages of design. Data collection

on usage trends, user happiness, and the overall user experience may be necessary for this. Overall, design research's function in the design process is to lay a strong foundation of understanding about the users and their demands, which can guide and inspire the creation of innovative and efficient solutions.

If we now go into detail about the different methods available in the area of interaction design research, we can see that there are several approaches to interaction design research, which can be broadly considered as fast or slow approaches. Fast approaches are characterized by their focus on rapid iteration and prototyping, often involving user testing at various stages of the design process. Slow approaches, on the other hand, tend to be more reflective and iterative, with a greater emphasis on understanding the context and needs of users over the long term. In this book, I explore both fast and slow approaches to interaction design research, examining their strengths and limitations and considering how they can be effectively combined in practice. In doing so, we also delve into the social science literature on topics such as user-centered design, usability engineering, and participatory design, as well as the specific field of human–computer interaction (HCI).

At the current moment, there is a growing "slow research" movement in HCI. This movement is not about "slow science," but rather about the development of a new sensibility that allows for alternative sets of research questions. As we slow down, we can start to see things differently, and it allows us to redirect our focus and our efforts. And as we change perspective from a design-oriented or solution-oriented approach that foregrounds how to quickly solve problems [sometimes referred to as "solutionism" (Cunningham et al., 2023)], we start to engage with questions that might not have any clear answers, simple solutions, or "quick fixes," for instance, ways of addressing social (un)justice, gender inequalities, and the global sustainability challenges. In short, it is a move toward "post-growth" approaches to HCI and interaction design research (Sharma et al., 2023).

Coming back to the definitions at the beginning of this chapter, I would in this context say that the "fast" vs. "slow" approaches to design research refer to the speed at which data is collected and analyzed, as well as the depth and breadth of the research – again in relation to the real-world research problem, and in relation to the research plan for the project. To provide an overview of these two approaches, one can say the following:

Fast design research methods are guidelines for those studies that can be conducted quickly and with minimal resources, and often involve

collecting data from a small number of sources. Examples of fast design research methods include online surveys, quick user interviews, rapid prototyping, and lightweight usability testing. These methods are useful for gathering initial data and testing basic assumptions, but may not provide as much depth or nuance as slower methods.

Slow design research methods, on the other hand, are guidelines for more time-consuming and resource-intensive studies, which often involve collecting data from a larger number of sources. Examples of slow design research methods include in-depth interviews, focus groups, and longitudinal field studies (Åhlström & Karlsson, 2010; Kjærup et al., 2021; Lucero et al., 2021). These methods can provide a more comprehensive understanding of the users and their needs but may take longer to conduct and analyze. For sure, if addressing systemic problems, there is a need for such in-depth studies.

In this book, I pinpoint how both fast and slow design research methods have their place in the design research process. Fast methods can be useful for gathering initial data and testing basic assumptions, while slow methods can provide a more comprehensive understanding of the users and their needs. In a sense, this sounds quite simple and straightforward. Still, combining these different approaches to make timely research contributions is a challenge. For instance, one might move too quickly and be too early, or one might be too slow and seek to make a contribution when similar projects have been conducted, or when similar solutions or results have already been published, or when the interest in a topic has already started to fade away. For anyone who seeks to make timely research contributions, this is something that needs to be carefully considered.

Accordingly, one of the key takeaways that will emerge across the following chapters of this book is the importance of considering ways of *combining fast and slow approaches* to ensure timely research contributions. Another important takeaway is the role of collaboration and cocreation in interaction design research. Both fast and slow approaches often involve collaboration with users, stakeholders, designers, and interdisciplinary research teams. By involving users in the design process and working in cross-disciplinary research teams, design researchers can gain valuable insights and feedback that could inform the design of more effective and satisfying interactions.

An additional reason for why it is important to be timely in terms of research interests, approach, and contributions is related to technological developments, changes in society, and changes in the research funding

landscape (which typically reflect advances in technologies and changing societal needs). For instance, there are currently several calls for research on sustainability or projects related to new technologies such as AI, Internet of Things (IoT), and cyber security – topics with clear societal relevance at this moment. At the same time, these topics are rather short term as societal needs change, and new technologies are constantly replacing other (older) technologies – just look at the technology trends over the last decade.

For sure, there are multiple reasons for digging into the temporalities at play in this research landscape, but before we move further into this topic, I should also say a few words about the overarching goals of this book.

About the Overarching Goals of This Book, before We Move on

There are several goals for this book. On an overarching level, I see it as a book that provides an overview of the different methods of design research, including both fast and slow approaches. Second, I see it as an additional goal to outline and describe the benefits and limitations of each method, and when it is most appropriate to use them.

In this book, I describe and show how fast and slow methods can be combined in an iterative design research process to gather data at different stages of the design process. I also offer practical examples, suggestions, techniques, and guidance for conducting interaction design research, including selecting the right methods, gathering and analyzing data, and presenting findings. Additionally, I also provide examples of how fast and slow methods have been used in real-world design projects to illustrate the practical application of these techniques. Finally, as a third goal, I hope this book will inspire designers to find and decide upon an appropriate and timely approach to interaction design research to create more effective, impactful designs as well as more long-lasting results.

In the following chapters, we will explore these themes in greater depth, examining the different methods and techniques used in fast and slow approaches to interaction design research. We will consider how they can be effectively combined to achieve optimal results. We also consider the challenges and ethical considerations that arise in this field and examine the latest trends and future directions in interaction design research.

As a start, the next chapter (Chapter 2) is devoted to "going fast," and it is accordingly about fast methods and approaches to interaction design research, and about how to be timely and relevant. After this chapter, I turn to ways of "going slow" in the research process (Chapter 3), including

not only the slow approach to research but also a discussion on long-lasting impact.

Overall, this book follows a reflective style of writing where we move from thinking about one approach to research, and how we then shift to a different approach. Hopefully, this reflective "thesis" – "antithesis" duality and structure – will help in moving forward to a synthesis on how these different approaches can be combined to produce both timely and well-grounded research in the area of HCI/interaction design research.

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